

Physical Chemistry (I)

Lecture 21

Chemical Equilibria

Electrochemistry



In the Last Class

- Colligative properties
 - Excess function
 - Activity
- Phase diagram of mixture
 - Ideal solutions
 - Real solutions: azeotropes
 - Real solutions: phase separation

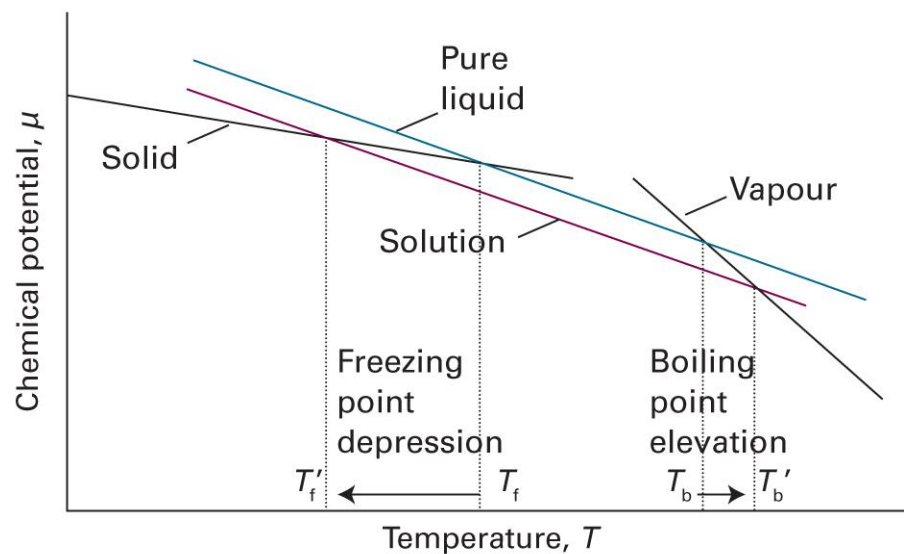
Colligative Properties

- Physical properties that depends on the relative number of solute particles present but not their chemical identity
1. Lowering of vapor pressure
 2. Elevation of boiling point
 3. Depression of freezing point
 4. Osmotic pressure

Last class

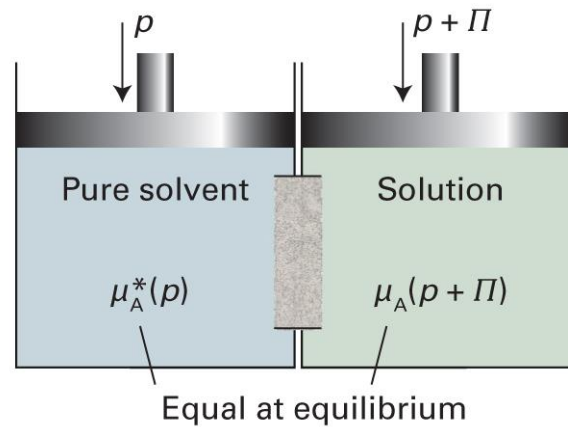
Changes in Phase Transitions

$$\mu_A = \mu_A^* + RT \ln x_A \quad (x_A < 1)$$



Last class

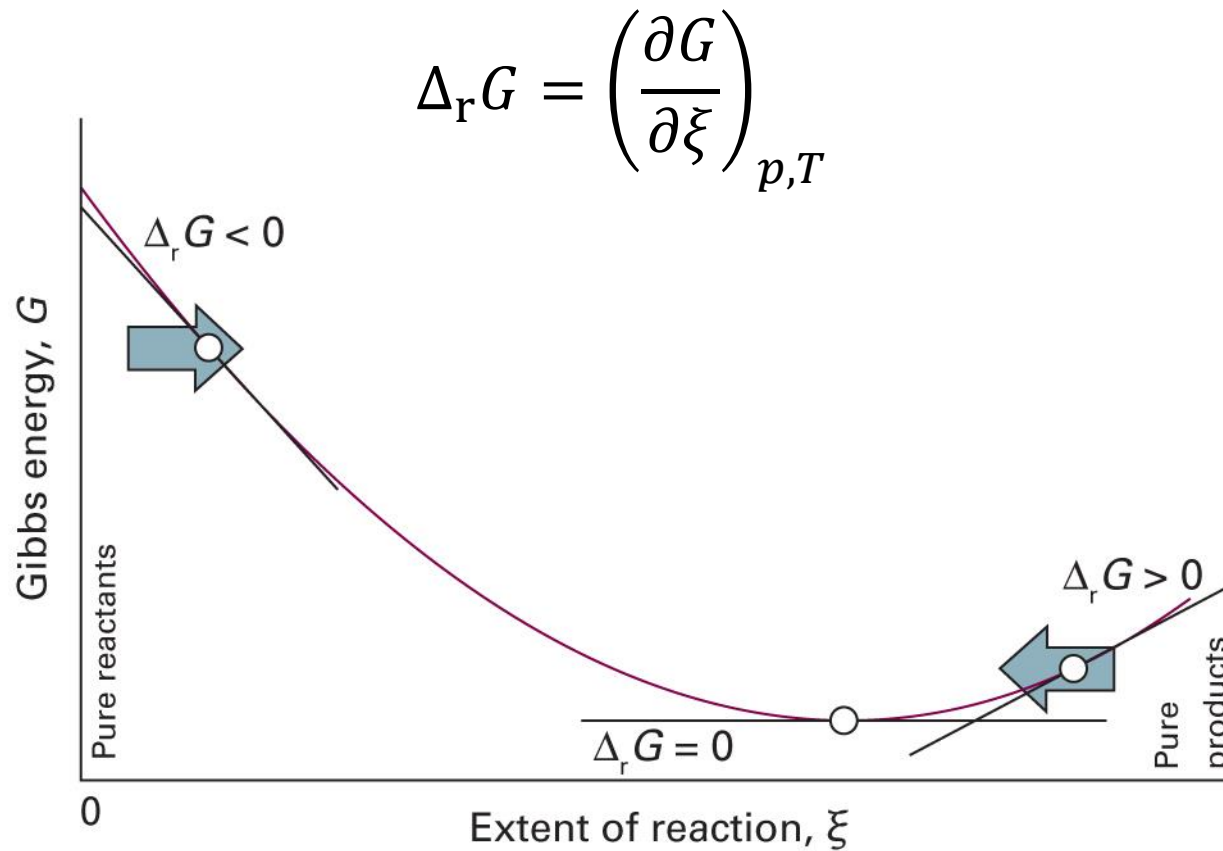
Thermodynamics of Osmosis



$$\mu_A^*(p) = \mu_A(x_A, p + \Pi)$$

$$\Pi = [B]RT \text{ (van't Hoff equation)}$$

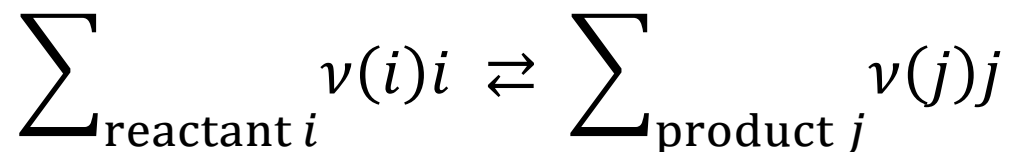
Reaction Gibbs Energy



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 6A.1)

Generalization

For an arbitrary reaction,



$$\Delta_r G = \left(\frac{\partial G}{\partial \xi} \right)_{p,T} = \sum_{\text{product } j} \mu_j \nu_j - \sum_{\text{reactant } i} \mu_i \nu_i$$

Chemical Potential and Activity

- Real gas: $\mu = \mu^\ominus + RT \ln(f/p^\ominus)$
- Real solution (solvent): $\mu = \mu^* + RT \ln a$
- Real solution (solute): $\mu = \mu^\ominus + RT \ln a$
- General: $\mu = \mu^\ominus + RT \ln a$

Component	Basis	Standard state	Activity	Limits
Solid or liquid		Pure, 1 bar	$a = 1$	
Solvent	Raoult	Pure solvent, 1 bar	$a = p/p^*, a = \gamma x$	$\gamma \rightarrow 1$ as $x \rightarrow 1$ (pure solvent)
Solute	Henry	(1) A hypothetical state of the pure solute	$a = p/K, a = \gamma x$	$\gamma \rightarrow 1$ as $x \rightarrow 0$
		(2) A hypothetical state of the solute at molality $b^\ominus$	$a = \gamma b/b^\ominus$	$\gamma \rightarrow 1$ as $b \rightarrow 0$
Gas	Fugacity [†]	Pure, a hypothetical state of 1 bar and behaving as a perfect gas	$f = \gamma p$	$\gamma \rightarrow 1$ as $p \rightarrow 0$



Generalization

$$\Delta_r G = \sum_{\text{product } j} \nu_j \mu_j - \sum_{\text{reactant } i} \nu_i \mu_i$$

For each species x ,

$$\nu_x \mu_x = \nu_x (\mu_x^\ominus + RT \ln a_x) = \nu_x \mu_x^\ominus + RT \ln(a_x^{\nu_x})$$

Then

$$\begin{aligned} \sum_x \nu_x \mu_x &= \sum_x \nu_x \mu_x^\ominus + RT \sum_x \ln(a_x^{\nu_x}) \\ &= \sum_x \nu_x \mu_x^\ominus + RT \ln \prod_x a_x^{\nu_x} \end{aligned}$$

Generalization

$$\Delta_r G = \sum_{\text{product } j} \nu_j \mu_j - \sum_{\text{reactant } i} \nu_i \mu_i$$

Substitution leads to

$$\Delta_r G = \Delta_r G^\ominus + RT \ln \prod_{\text{product } j} a_j^{\nu_j} - RT \ln \prod_{\text{reactant } i} a_i^{\nu_i}$$

where $\Delta_r G^\ominus = \sum_{\text{product } j} \nu_j \mu_j^\ominus - \sum_{\text{reactant } i} \nu_i \mu_i^\ominus$

$$\Delta_r G = \Delta_r G^\ominus + RT \ln \left(\prod_{\text{product } j} a_j^{\nu_j} / \prod_{\text{reactant } i} a_i^{\nu_i} \right)$$

Q and K

$$\Delta_r G = \Delta_r G^\ominus + RT \ln Q$$

where the **reaction quotient** Q is defined as

$$Q = \frac{\prod_{\text{product } j} a_j^{v_j}}{\prod_{\text{reactant } i} a_i^{v_i}}$$

At equilibrium, $\Delta_r G = 0$

$$0 = \Delta_r G^\ominus + RT \ln Q_{\text{eq}} = \Delta_r G^\ominus + RT \ln K$$

where the **equilibrium constant** K is Q at equilibrium

Connection to Thermochemistry

$$\Delta_r G^\ominus = -RT \ln K$$

Since we know

$$\Delta_r G^\ominus = \sum_{\text{products } j} \nu_j \Delta_f G^\ominus(j) - \sum_{\text{reactants } i} \nu_i \Delta_f G^\ominus(i),$$

we can calculate K from thermochemical data.

Activity Coefficients

For the solution-phase reaction $A + B \rightleftharpoons C + D$,

$$K = \frac{a_C a_D}{a_A a_B} = \frac{(\gamma_C [C]/c^\ominus)(\gamma_D [D]/c^\ominus)}{(\gamma_A [A]/c^\ominus)(\gamma_B [B]/c^\ominus)}$$

Typically, we assume $\gamma_i \approx 1$ to approximate

$$K \approx \frac{([C]/c^\ominus)([D]/c^\ominus)}{([A]/c^\ominus)([B]/c^\ominus)}$$

Approximation for Activity

State	Measure	Approximation for a_j	Definition
Solute	molality	$b_j/b^\ominus$	$b^\ominus = 1 \text{ mol kg}^{-1}$
	molar concentration	$[J]/c^\ominus$	$c^\ominus = 1 \text{ mol dm}^{-3}$
Gas phase	partial pressure	$p_j/p^\ominus$	$p^\ominus = 1 \text{ bar}$
Pure solid, liquid		1 (exact)	

Pure Solids and Liquids

$$\Delta G_{\text{m}} = \mu_f - \mu_i = \int_{p_i}^{p_f} V_{\text{m}} \, dp$$

Let the initial state to be the standard state:

$$\mu - \mu^{\ominus} = \int_{p^{\ominus}}^p V_{\text{m}} \, dp' \approx V_{\text{m}} \int_{p^{\ominus}}^p 1 \, dp'$$

From the definition of an activity, $\mu - \mu^{\ominus} = RT \ln a$,

$$\ln a = \frac{V_{\text{m}}}{RT} \int_{p^{\ominus}}^p 1 \, dp' = \frac{V_{\text{m}}}{RT} (p - p^{\ominus})$$

Not in Atkins

Example: Activity of Coke

The density of coke at 1000°C is about 1.5 g cm⁻³. Calculate the activity of C(s) in the form of coke at 100 bar and 1000°C.

The molar volume is

$$V_m = (\text{molar mass})/(\text{density}) = (12 \text{ g/mol})/(1.5 \text{ g cm}^{-3}) = 8.0 \text{ cm}^3 \text{ mol}^{-1}$$

Thus,

$$\begin{aligned}\ln a &= V_m(p - p^\ominus)/RT \\ &= (8.0 \text{ cm}^3 \text{ mol}^{-1}) \times (100 - 1) \text{ bar} / (0.082 \text{ L bar K}^{-1} \text{ mol}^{-1}) / (1273 \text{ K}) \\ &= 0.0076\end{aligned}$$

or $a = 1.01$.



Pure Solids and Liquids

$$\ln a = \frac{V_m}{RT} (p - p^\ominus)$$

At moderate pressures, the activity is very close to 1.

Hence, we usually take $a = 1$ for pure solids and liquids.

For Gaseous Reactions

$$K = \frac{\prod_{\text{product } j} a_j^{\nu_j}}{\prod_{\text{reactant } i} a_i^{\nu_i}} = \frac{\prod_{\text{product } j} (p_j/p^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (p_i/p^\ominus)^{\nu_i}}$$

Since $p_i/p^\ominus = [i](RT/p^\ominus) = [i]/c^\ominus (c^\ominus RT/p^\ominus)$,

$$K = \frac{\prod_{\text{product } j} ([j]/c^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} ([i]/c^\ominus)^{\nu_i}} \left(\frac{c^\ominus RT}{p^\ominus} \right)^{\sum_j \nu_j - \sum_i \nu_i}$$

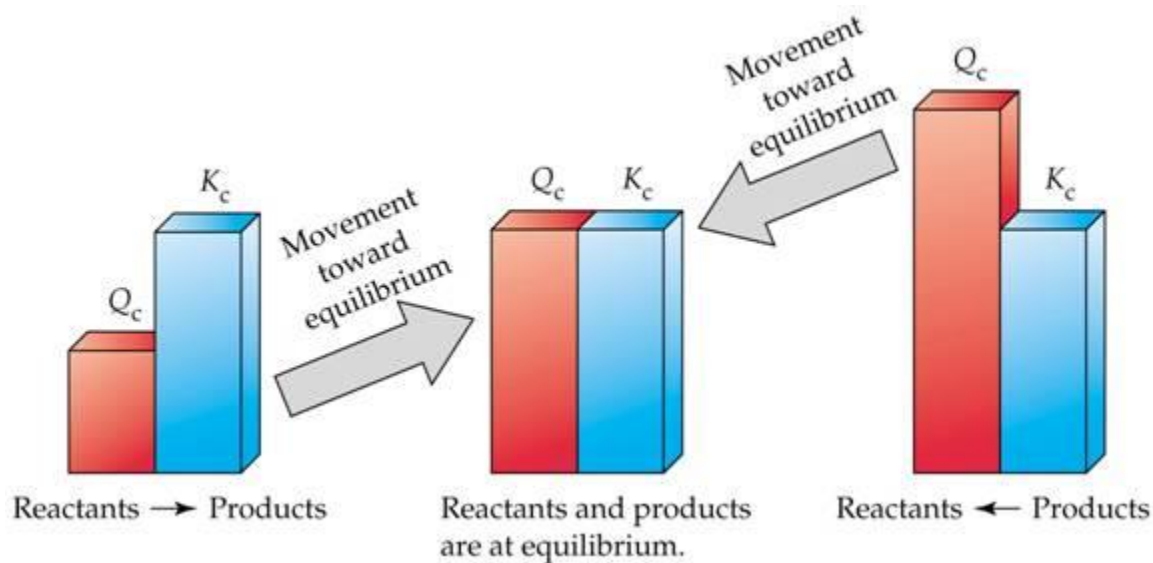
$$K = K_c \left(\frac{c^\ominus RT}{p^\ominus} \right)^{\Delta \nu} = K_c \left(\frac{T}{12.03 \text{ K}} \right)^{\Delta \nu}$$

Meaning of Q and K

- Reaction quotient Q
 - “How far the equilibrium is”
 - Varies as the reaction progresses
 - $Q = K$ at equilibrium
- Equilibrium constant K
 - Amounts of reactants and products at equilibrium
 - Constant for the given experimental condition

Not in Atkins

Meaning of Q and K



Le Chatelier's Principle

When subjected to a disturbance,
a system at equilibrium tends to respond in a way
that **minimizes the effect of the disturbance**.

Disturbances to consider:

- Amount (concentration, pressure)
- Temperature

Amount Change

- **Increase** of the amount of a particular species
 - The system responds to **decrease** the amount
- **Decrease** of the amount of a particular species
 - The system responds to **increase** the amount
- Reactant increase / product decrease: $Q < K$
 - Decrease of reactants & increase of products
- Reactant decrease / product increase: $Q > K$
 - Increase of reactants & decrease of products

Volume Change

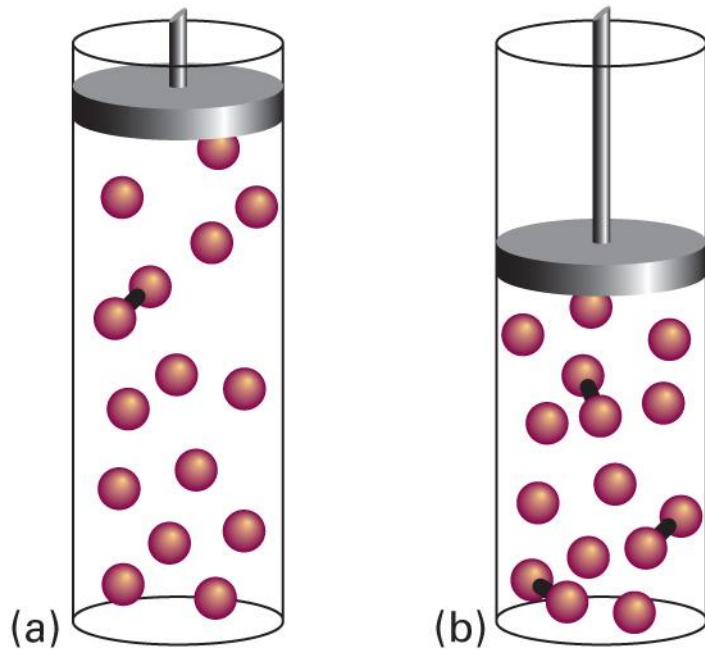
For a gaseous reaction,

- Increase of the volume → decrease of the total amount
- Decrease of the volume → increase of the total amount

(Think of concentrations)

Pressure change is linked to volume change via $pV = nRT$

Volume Change



Volume Change

$$K_c = \frac{\prod_{\text{product } j} ([j]/c^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} ([i]/c^\ominus)^{\nu_i}} = \frac{\prod_{\text{product } j} (n_j/Vc^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (n_i/Vc^\ominus)^{\nu_i}}$$

For a volume change from V to xV ,

$$Q_c = \frac{\prod_{\text{product } j} (n_j/xVc^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (n_i/xVc^\ominus)^{\nu_i}} = K_c x^{-(\sum_j \nu_j - \sum_i \nu_i)} = K_c x^{-\Delta \nu}$$

Volume Change

$$Q_c = K_c x^{-\Delta\nu}$$

For $x > 1$ (volume increase),

- $\Delta\nu = 0$: $Q_c = K_c$ and the system is at equilibrium
- $\Delta\nu > 0$: $Q_c < K_c$ and the forward reaction is favored
- $\Delta\nu < 0$: $Q_c > K_c$ and the reverse reaction is favored

Temperature Change

- Exothermic reactions
 - Increased temperature favors the reactants
 - Decreased temperature favors the products
- Endothermic reactions
 - Increased temperature favors the products
 - Decreased temperature favors the reactants

van't Hoff Equation

$$\ln K = -\Delta_r G^\ominus / RT$$

Differentiating both sides with respect to T ,

$$\frac{d \ln K}{dT} = -\frac{1}{R} \frac{d(\Delta_r G^\ominus / T)}{dT}$$

Using the Gibbs-Helmholtz equation,

$$-R \frac{d \ln K}{dT} = \frac{d(\Delta_r G^\ominus / T)}{dT} = -\frac{\Delta_r H^\ominus}{T^2}$$

$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\ominus}{RT^2}$$

T*-Dependence of *K

$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\ominus}{RT^2}$$

Integration gives

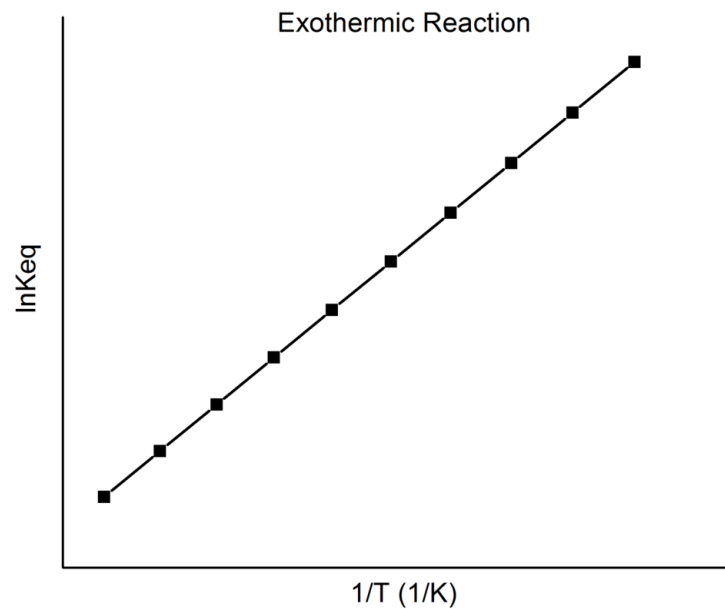
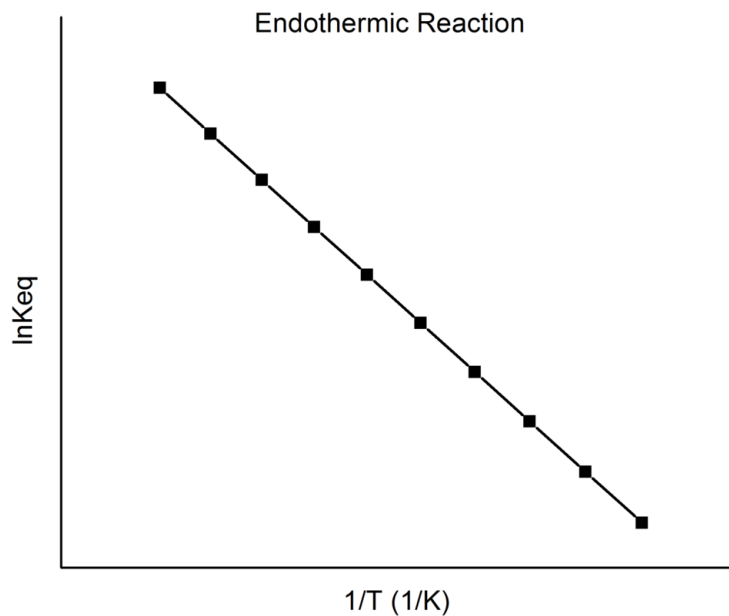
$$\int_{\ln K_1}^{\ln K_2} d \ln K = \int_{T_1}^{T_2} \frac{\Delta_r H^\ominus}{RT^2} dT$$

Assuming that $\Delta_r H^\ominus$ is constant,

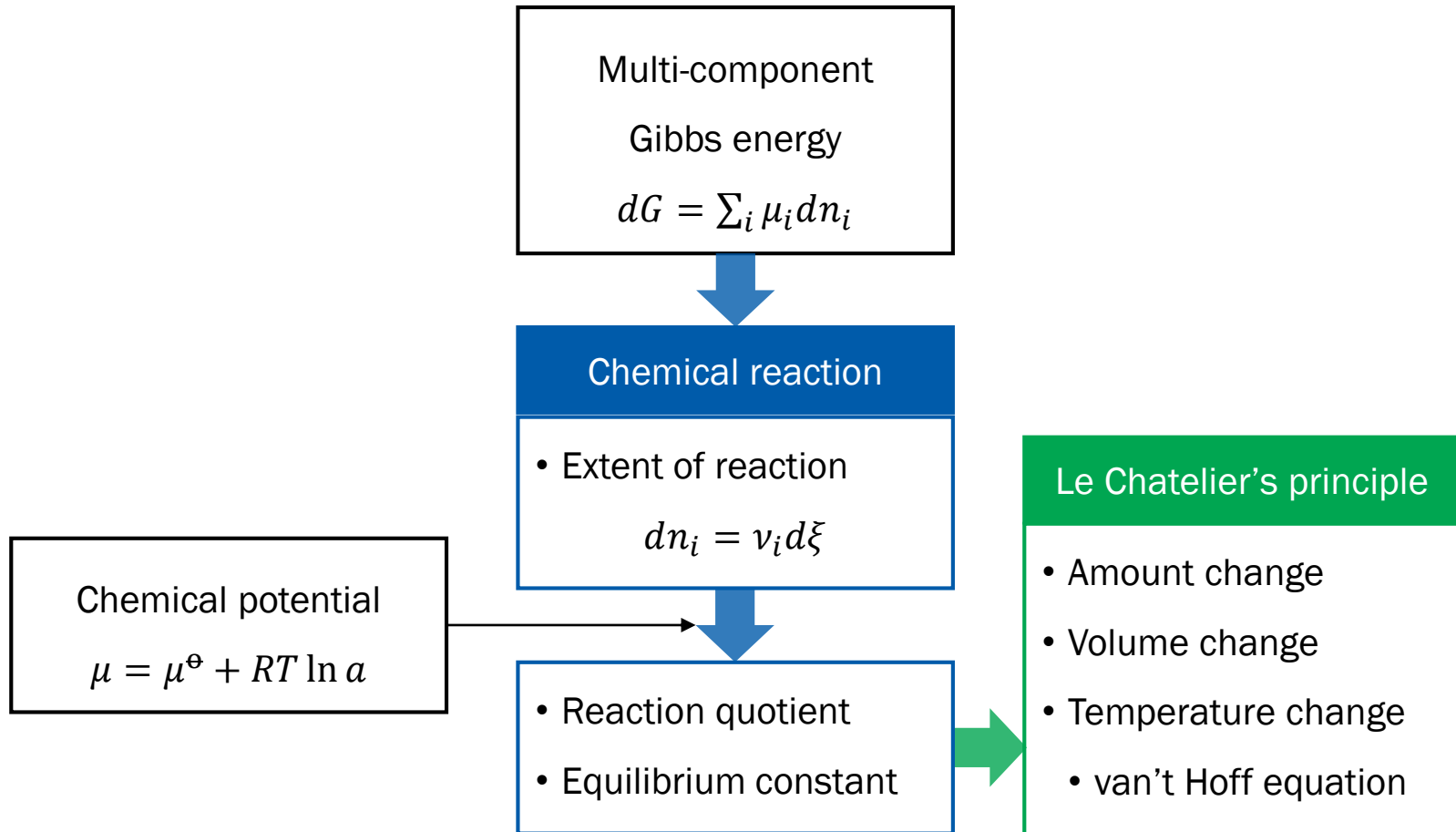
$$\ln K_2 - \ln K_1 = \frac{\Delta_r H^\ominus}{R} \int_{T_1}^{T_2} \frac{1}{T^2} dT = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

T-Dependence of *K*

$$\ln K_2 - \ln K_1 = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$



Summary



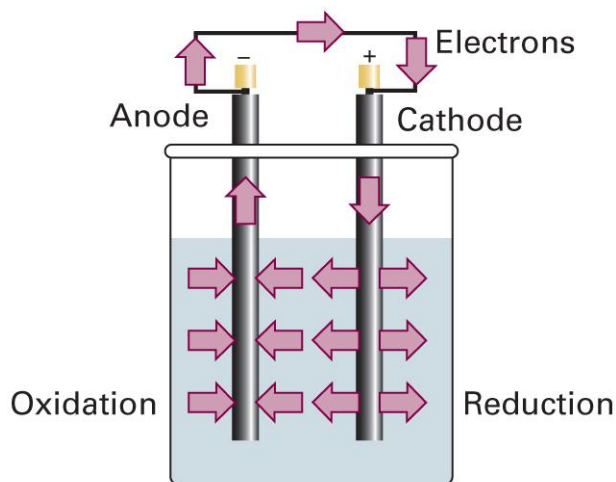
9. Electrochemistry

Chapter Overview

- Electrochemical cells
 - Basic terminology and concepts
- Thermodynamics of cells
 - Cell potential and reaction
 - Electrode potentials

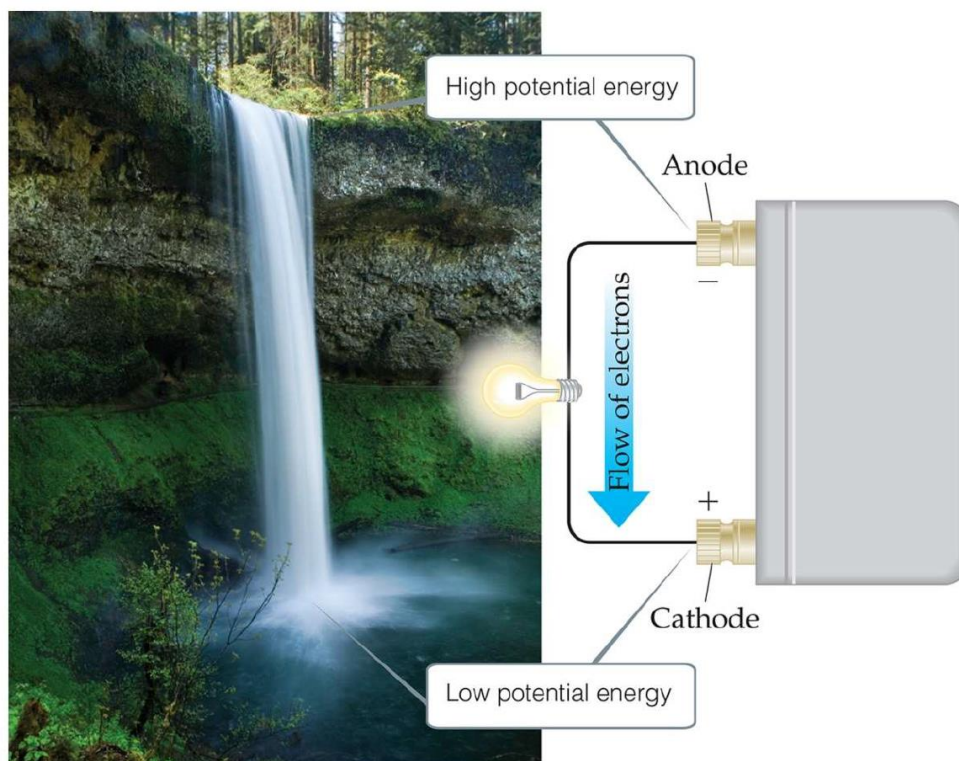
Electrochemical Cell

- Two electrodes + electrolytes
 - Anode: oxidation
 - Cathode: reduction



Cell Potential

Potential difference between the two electrodes



(Image: <https://www.unf.edu/~michael.lufaso/chem2046/2046chapter20.pdf>)

Thermodynamics of Cell

Recall that

$$dG_{T,p} = \delta w_{\text{add,max}} = \delta w_{\text{elec}}$$

for a **reversible** electrochemical reaction.

For an infinitesimal progress of the reaction $d\xi$,

$$dG = \Delta_r G d\xi$$

Hence,

$$\delta w_{\text{elec}} = \Delta_r G d\xi$$

Thermodynamics of Cell

$$\delta w_{\text{elec}} = \Delta_r G d\xi$$

On the other hand, the electrical work is

$$\delta w_{\text{elec}} = \phi dQ$$

where ϕ is the electric field and dQ is the transferred charge. For the electrochemical reaction,

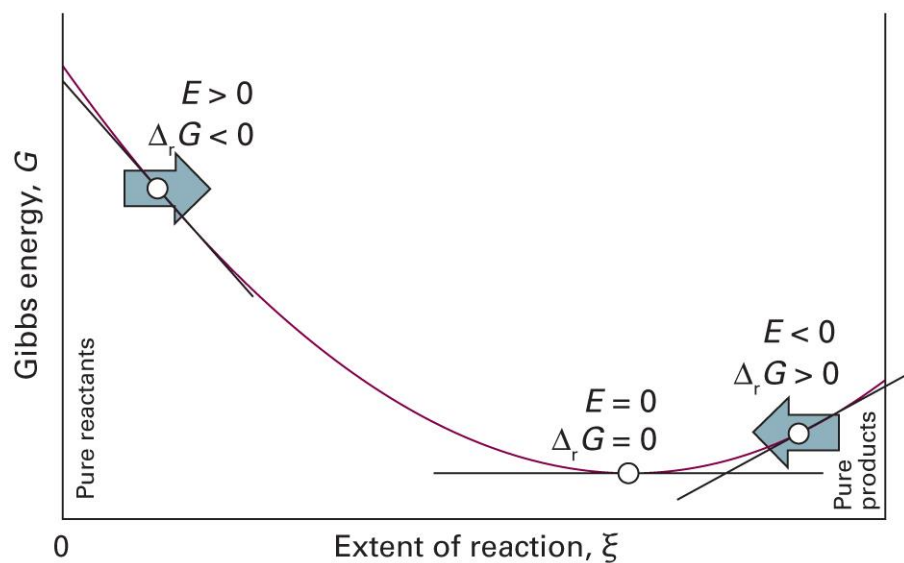
$$\phi = E_{\text{cell}}, \quad dQ = -\nu e N_A d\xi = -\nu F d\xi$$

$$\delta w_{\text{elec}} = -\nu F E_{\text{cell}} d\xi$$

Thus, $\Delta_r G = -\nu F E_{\text{cell}}$

Cell Potential and Reaction

$$\Delta_r G = -\nu F E_{\text{cell}}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 6C.4)

Nernst Equation

$$\Delta_r G = -\nu F E_{\text{cell}}$$

Using $\Delta_r G = \Delta_r G^\ominus + RT \ln Q$,

$$-\nu F E_{\text{cell}} = \Delta_r G^\ominus + RT \ln Q$$

Thus,

$$E_{\text{cell}} = E_{\text{cell}}^\ominus - \frac{RT}{\nu F} \ln Q$$

where $E_{\text{cell}}^\ominus = -\Delta_r G^\ominus / \nu F$

At Equilibrium

$$E_{\text{cell}} = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln Q$$

At equilibrium, $E_{\text{cell}} = 0$ and $Q = K$

$$0 = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln K$$

or,

$$E_{\text{cell}}^{\ominus} = \frac{RT}{\nu F} \ln K$$

Temperature Coefficient

$$E_{\text{cell}}^{\ominus} = -\frac{\Delta_{\text{r}}G^{\ominus}}{\nu F}$$

Since $(\partial G / \partial T)_p = -S$,

$$\frac{dE_{\text{cell}}^{\ominus}}{dT} = -\frac{1}{\nu F} \left(\frac{\partial \Delta_{\text{r}}G^{\ominus}}{\partial T} \right)_p = \frac{\Delta_{\text{r}}S^{\ominus}}{\nu F}$$

Furthermore,

$$\Delta_{\text{r}}H^{\ominus} = \Delta_{\text{r}}G^{\ominus} + T\Delta_{\text{r}}S^{\ominus} = -\nu F \left(E_{\text{cell}}^{\ominus} - T \frac{dE_{\text{cell}}^{\ominus}}{dT} \right)$$

Electrode Potentials

Cell potential = cathode contribution + anode contribution

For a cell L||R,

$$E_{\text{cell}}^{\ominus} = E^{\ominus}(\text{R}) - E^{\ominus}(\text{L})$$

Impossible to measure each contribution

Convention

- Standard hydrogen electrode (SHE): $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{H}^+(\text{aq})$

$E^\ominus = 0$ at all temperatures

- Standard potential $E^\ominus(\text{X})$

For a cell $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{H}^+(\text{aq}) || \text{X}$, $E^\ominus(\text{X}) = E_{\text{cell}}^\ominus$

Standard Potentials

Couple	$E^\ominus/\text{V}$
$\text{Ce}^{4+}(\text{aq}) + \text{e}^- \rightarrow \text{Ce}^{3+}(\text{aq})$	+1.61
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$	+0.34
$\text{AgCl}(\text{s}) + \text{e}^- \rightarrow \text{Ag}(\text{s}) + \text{Cl}^-(\text{aq})$	+0.22
$\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \frac{1}{2}\text{H}_2(\text{g})$	0
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	-0.76
$\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$	-2.71

Thermodynamics of Potentials

Each electrode potential can be converted into

$$-vFE^{\ominus} = \Delta_r G^{\ominus}$$

When a combination of reactions a and b gives reaction c,

$$\Delta_r G^{\ominus}(c) = \Delta_r G^{\ominus}(a) + \Delta_r G^{\ominus}(b)$$

$$-v_c FE^{\ominus}(c) = -v_a FE^{\ominus}(a) - v_b FE^{\ominus}(b)$$

Thus,

$$v_c E^{\ominus}(c) = v_a E^{\ominus}(a) + v_b E^{\ominus}(b)$$

Reducing Power

For two redox couples $\text{Ox}_\text{L}/\text{Red}_\text{L}$ and $\text{Ox}_\text{R}/\text{Red}_\text{R}$,
the cell reaction $\text{Red}_\text{L} + \text{Ox}_\text{R} \rightarrow \text{Ox}_\text{L} + \text{Red}_\text{R}$ has

- $K > 1$: $E^\ominus_{\text{cell}} > 0$ and $E^\ominus(\text{L}) < E^\ominus(\text{R})$
- $K < 1$: $E^\ominus_{\text{cell}} < 0$ and $E^\ominus(\text{L}) > E^\ominus(\text{R})$

Red_L has a thermodynamic tendency to reduce Ox_R
if $E^\ominus(\text{L}) < E^\ominus(\text{R})$

Electrochemical Series

Least strongly reducing

Gold (Au^{3+}/Au)

Platinum (Pt^{2+}/Pt)

Silver (Ag^+/Ag)

Mercury (Hg^{2+}/Hg)

Copper (Cu^{2+}/Cu)

Hydrogen (H^+/H_2)

Tin (Sn^{2+}/Sn)

Nickel (Ni^{2+}/Ni)

Iron (Fe^{2+}/Fe)

Zinc (Zn^{2+}/Zn)

Chromium (Cr^{3+}/Cr)

Aluminium (Al^{3+}/Al)

Magnesium (Mg^{2+}/Mg)

Sodium (Na^+/Na)

Calcium (Ca^{2+}/Ca)

Potassium (K^+/K)

Most strongly reducing

